

National Land Acquisition & Management System (NLAMS)

Smart India Hackathon (SIH) 2026

Web3 & GIS Enabled End-to-End Monitoring Portal Report

PROJECT METADATA

Problem Statement:	Real-Time National Land Acquisition & Management System
Focus Areas:	Web3 Blockchain, GIS Parcel Mapping, LARR Workflow Router
Prototype Version:	v2.0 (Modernized Government Light Theme Portal)
Compiled Date:	August 2026

1. Overall Website Layout & Page Features

The NLAMS system translates the LARR (Land Acquisition Act, 2013) guidelines into a unified digital pipeline. The pages serve the following purposes:

Home / Landing Portal

Styled as an official national entry portal, featuring a tricolor header band, a hero overview of the LARR roadmap, and animated highlights of the system's operational pillars.

Executive Analytics & Monitoring (Dashboard)

Designed for central and state decision-makers. Features:

- **Dynamic KPIs:** Tracks total notified vs acquired land area, compensation disbursement ratios, and average resettlement rates.
- **GIS Visualizer:** Renders land parcel coordinates as interactive map polygons. Colors update dynamically based on stage.
- **Mitigation Risk Index:** Provides risk alerts, completion forecasts, and bottleneck analysis based on regional parcel fragmentation and current step latency.
- **MIS Export:** An active button compiling project metrics into a downloadable CSV spreadsheet on the fly.

Proposals & Automated Workflows

The operational pipeline. Central ministries can register new proposals. State, district, and surveyor profiles can switch roles in the header and verify/approve subsequent milestones. Completed stages feature printable gazette document buttons.

Web3 Trust & Compensation Portal

The decentralised backend simulation. Integrates a mock Metamask connection to disburse escrow funds. Logs every action (payouts, titles, geo-fences) in a state audit block ledger. Includes a digital deed upload widget that computes file SHA-256 hashes to verify authenticity.

Field Surveyor Node

Mobile-responsive field interface. Simulates GPS sensor triangulation to capture boundary points. Records valuation attributes (soil fertility, structure types) and uploads site photos.

2. Role-Based Access Control (RBAC) & Simulations

The NLAMS system features role-based access control to reflect the multiple authorities involved in land acquisition. Switching user profiles in the portal alters the available actions:

Central Ministry (ministry)

Has super-admin capabilities to register new land acquisition proposals, view national statistics, and sign off on any workflow stage for ease of demonstration.

State Government (state)

Responsible for Stage 2: GIS Verification. When a proposal is submitted, only the State Government profile can verify boundary coordinates and record spatial alignment hashes on the ledger.

District Authority / Collector (district)

Responsible for Stage 3 (Section 11 Notification) and Stage 4 (Award Declaration). Collector actions publish legal intent and lock valuation budgets in the escrow smart contracts.

Field Surveyor (surveyor)

Granted access to the Field Surveyor Node to trigger mobile-responsive GPS sensor detections, classify soil quality, inspect unbuilt plots, and sign off on Stage 5: Possession Handover.

Citizen / Land Owner (citizen)

View-only access for workflows. Permitted to connect a crypto wallet in the Compensation Portal to claim payments directly from the escrow account, keeping transactions completely transparent.

3. Tech Stack Breakdown

The application is developed using standard industry-level technologies suited for fast hot-reloading and polished UI output under pressure:

- **Frontend Framework:** React.js built on Vite (Vite 8, React 19) for immediate loading times and HMR (Hot Module Replacement).
- **Styling Stack:** Tailwind CSS v4 for fully custom responsive components, clean government-themed color palettes, and layouts.
- **GIS Map Rendering:** React Leaflet & Leaflet.js utilizing OpenStreetMap tile layers to draw and render coordinate boundaries (polygons).
- **MIS Visual Charts:** Recharts package using SVG elements for responsive financial and progress bar graphs.
- **Icons:** Lucide React icons for clean, minimalistic government and technical emblems.
- **Web3 Simulation:** Context-based mock ledger engine simulating SHA-256 hashes, transaction block creation, and crypto signatures.

4. Path to Production: Making the Project Real

To elevate this frontend prototype into a production-grade system for the final hackathon round, the following back-end integrations are required:

A. Smart Contract Deployments (Web3)

Replace the mockup ledger with actual Solidity Smart Contracts deployed to an Ethereum Layer 2 (e.g. Arbitrum, Optimism) or Polygon. Use contracts to tokenize land titles as Soulbound NFTs (non-transferable title deeds) and lock compensation in Escrow wallets that auto-release on PFMS verification.

B. Real API Integrations

Connect actual government databases to feed data into the portal:

- **BHOOMI / State Registries:** Connect REST APIs to fetch official land ownership details, preventing double-selling disputes.
- **BhuNaksha:** Import XML/GeoJSON cadastral maps to verify physical land shapes on the map.
- **PFMS Gateway:** Integrate the Public Financial Management System to trigger real-time bank transfers.

C. Geospatial Database

Use a database like PostgreSQL with the PostGIS extension. This allows you to run spatial SQL queries (e.g. detect if highway path intersects with protected forest reserves or existing private buildings).

D. Official Authentication

Implement Single Sign-On (SSO) using India's Digilocker or Aadhaar API, verifying the identities of surveyors, district magistrates, and farmers.